

PROJECT DESIGN PHASE

PROBLEM–SOLUTION FIT

Date	3rd October 2026
Team ID	6ab8c0aa8c7c66e70673078c
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	

4. PROJECT DESIGN

The Project Design Phase converts the requirements identified during the Requirement Analysis Phase into a structured technical solution. For the Script-Controlled ACL – Restrict Record Access Based on Field Value project, the design focuses on how ServiceNow ACLs, custom roles, field conditions, and the Institution Details table work together to provide controlled record access.

4.1 PROBLEM–SOLUTION FIT

IDENTIFIED PROBLEM

The Institution Details (u_institution_details) table contains student records belonging to different branches such as ECE, EEE, and CSE. Users should not have unrestricted access to all records or all record operations.

The system requires:

- Record-level access control.
- Branch-based restriction for Read access.
- Different permissions for different users.
- Separate control over Read, Create, Write, and Delete operations.
- Prevention of unauthorized operations.

A basic table-level permission mechanism is not sufficient because the project requires the system to consider both the user's assigned role and the value of the Branch field.

PROPOSED PROBLEM–SOLUTION FIT

The project addresses these requirements using ServiceNow Access Control Lists (ACLs) combined with custom roles and a field-based condition.

Identified Problem	Proposed Solution
Unauthorized users may access restricted records	Use ServiceNow ACLs to control record access
Users require different levels of access	Create custom roles bb1, bb2, bb3, and bb4
Records must be restricted based on branch	Use the Branch field as an access condition
Only permitted EEE records should be readable	Configure the READ ACL with the required role and Branch = EEE condition
Record creation must be controlled	Configure a CREATE ACL requiring bb2
Record modification must be controlled	Configure a WRITE ACL requiring bb3
Record deletion must be controlled	Configure a DELETE ACL requiring bb4
Security configuration must be verified	Test different users and role combinations

PROBLEM–SOLUTION RELATIONSHIP